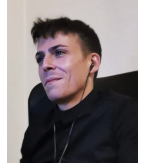


CURRICULUM VITAE

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 Position: Research scientist, TRIPOP-LJK, Inria Centre at Université Grenoble Alpes
 Profiles: [Scholar](#) – [ORCID](#) – [Scopus](#) – [GitHub](#) – [LinkedIn](#) – [ResearchGate](#)



1 Employment

01/2025 – present Research scientist, TRIPOP project-team, Laboratoire Jean Kuntzmann (LJK), Inria Centre at Université Grenoble Alpes, France.
 01/12/2022 – 12/2024 Postdoctoral Research Associate, Sydney Centre in Geomechanics and Mining Materials, School of Civil Engineering, The University of Sydney, Australia; ARC Discovery Project DP220101164.
 01/2021 – 06/2023 Contractual teacher, École Centrale de Nantes, France.
 01/2021 – 11/2022 Postdoctoral researcher under ERC-StG CoQuake – Grant agreement ID 757848, GeM (UMR 6183), École Centrale de Nantes, France.
 11/2020 – 12/2020 Research Engineer, École Centrale de Nantes (Centrale Innovation), France.
 09/2017 – 10/2020 Engineer – CIFRE, Ingérop Conseil et Ingénierie, Rueil-Malmaison, France.

2 Academic degrees

10/2017 – 12/2020 PhD, Mechanics – École Centrale de Nantes (École des Ponts and Ingérop), France. Supervisors: I Stefanou, P Vannucci.
 01/2016 – 07/2017 Master, Mechanical Engineering, University of Florence, Italy. Supervisors: P M Mariano, B Facchini.
 10/2013 – 12/2015 Bachelor, Mechanical Engineering, University of Florence, Italy. Supervisor: P M Mariano.

3 Honours and awards

2022 [Early Career Researcher Award by EUROMECH \(European Mechanics Society\)](#) on the occasion of the 18th European Mechanics of Materials Conference, Oxford, UK.
 2021 [Award for the best PhD thesis](#) bringing technological and conceptual breakthroughs in the industry by Centrale Innovation (Écoles Centrales Group).
 2021 [Award](#) for the best PhD by the Computational Structural Mechanics Association (CSMA).

4 Research grants

2025–2029 Principal investigator (PI), MIAI Cluster IA Chair AIM: Artificial Intelligence and Mechanics for scale bridging in complex materials ([website](#)), with V Acary (Inria/LJK, PI). Funded by MIAI Cluster and ANR through the France 2030 programme (grant agreement ANR-23-IAEL-0006). Grant: 500 kEUR.
 2027–2031 Scientific leader (co-PI), ANR PRC MML-Geo: Mechanics-based Machine Learning for scale bridging in Geomechanics, with L-M Guayacán-Carrillo (Navier/ENPC, coordinator, PI), A Wautier (INRAe, co-PI), and B Pardoën (LTDS/ENTPE, co-PI). Grant: 700 kEUR.
 2027–2031 Associate member, ANR PRCE MODEMSEA: Numerical modeling of coastal defense structures using a fully coupled SPH-DEM-FEM model, with N-S Nguyen (GeM/ECN, coordinator, PI), O Goury (Inria/LJK, PI), P A Bouttier (GRICAD/UGA, PI), and O Bertrand (ARTELIA). Grant: 450 kEUR.

5 Scientific production

Author of 16 refereed journal articles and two book chapters, with additional reviewed conference contributions, communications/posters, and invited lectures and seminars.

5.1 Refereed journal articles

- [1] **F Masi**. “Learning inelastic constitutive models from stress–strain data under hard thermodynamic constraints”. In: *Computer Methods in Applied Mechanics and Engineering* 1.1 (2026), p. 1. DOI: <https://doi.org/10.1016/j.cma.2026.119260>.
- [2] **F Masi** and I Einav. “Hydrodynamics of fault gouges from constitutive modelling to the physics of friction”. In: *Journal of Geophysical Research: Solid Earth* 130.7 (2025), e2024JB030822. DOI: <https://doi.org/10.1029/2024JB030822>.
- [3] A Morsel, **F Masi**, P Kotronis, and I Stefanou. “Measurement, self-similarity, and TNT equivalence of blasts from exploding wires”. In: *Shock Waves* 35 (2025), pp. 17–35. DOI: [10.1007/s00193-024-01209-x](https://doi.org/10.1007/s00193-024-01209-x).
- [4] A Morsel, **F Masi**, E Marché, G Racineux, P Kotronis, and I Stefanou. “miniBLAST: a novel experimental setup for laboratory testing of structures under blast loads”. In: *Experimental Techniques* (in press) (2025). DOI: [10.1007/s40799-024-00771-4](https://doi.org/10.1007/s40799-024-00771-4).
- [5] **F Masi** and I Einav. “Neural integration for constitutive equations using small data”. In: *Comput Methods Appl Mech Eng* 420 (2024), p. 116698. DOI: [10.1016/j.cma.2023.116698](https://doi.org/10.1016/j.cma.2023.116698).
- [6] **F Masi** and I Stefanou. “[Evolution TANN and the identification of internal variables and evolution equations in solid mechanics](#)”. In: *J Mech Phys Solids* 174 (2023). DOI: [10.1016/j.jmps.2023.105245](https://doi.org/10.1016/j.jmps.2023.105245).

- [7] **F Masi** and I Stefanou. “Multiscale modeling of inelastic materials with Thermodynamics-based Artificial Neural Networks (TANN)”. In: *Comput Methods Appl Mech Eng* 398 (2022), p. 115190. DOI: [10.1016/j.cma.2022.115190](https://doi.org/10.1016/j.cma.2022.115190).
- [8] **F Masi**, I Stefanou, P Vannucci, and V Maffi-Berthier. “Thermodynamics-based Artificial Neural Networks for constitutive modeling”. In: *J Mech Phys Solids* 147 (2021), p. 104277. DOI: [10.1016/j.jmps.2020.104277](https://doi.org/10.1016/j.jmps.2020.104277).
- [9] **F Masi**, I Stefanou, and V Maffi-Berthier. “Scaling laws for rigid-body response of masonry structures under blast loads”. In: *J Eng Mech (featured in the Editor's Choice section)* 147.10 (2021), p. 04021078. DOI: [10.1061/\(ASCE\)EM.1943-7889.0001986](https://doi.org/10.1061/(ASCE)EM.1943-7889.0001986).
- [10] **F Masi**, I Stefanou, P Vannucci, and V Maffi-Berthier. “Resistance of museum artefacts against blast loading”. In: *J Cul Her* 44 (2020), pp. 163–173. DOI: [10.1016/j.culher.2020.01.015](https://doi.org/10.1016/j.culher.2020.01.015).
- [11] **F Masi**, I Stefanou, V Maffi-Berthier, and P Vannucci. “A Discrete Element Method based-approach for arched masonry structures under blast loads”. In: *Eng Struct* 216 (2020), p. 110721. DOI: [10.1016/j.engstruct.2020.110721](https://doi.org/10.1016/j.engstruct.2020.110721).
- [12] **F Masi**, I Stefanou, P Vannucci, and V Maffi-Berthier. “Rocking response of inverted pendulum structures under blast loading”. In: *Int J Mech Sci* 157-158 (2019), pp. 833–848. DOI: [10.1016/j.ijmecsci.2019.05.024](https://doi.org/10.1016/j.ijmecsci.2019.05.024).
- [13] P Vannucci, **F Masi**, and I Stefanou. “A nonlinear approach to the wind strength of Gothic Cathedrals: The case of Notre Dame of Paris”. In: *Eng Struct* 183 (2019), pp. 860–873. ISSN: 0141-0296. DOI: [10.1016/j.engstruct.2019.01.030](https://doi.org/10.1016/j.engstruct.2019.01.030).
- [14] **F Masi**, PM Mariano, and P Vannucci. “Blast actions in aircrafts: An integrated methodology for designing protection devices”. In: *Eng Struct* 175 (2018), pp. 895–911. DOI: [10.1016/j.engstruct.2018.08.082](https://doi.org/10.1016/j.engstruct.2018.08.082).
- [15] **F Masi**, I Stefanou, and P Vannucci. “A study on the effects of an explosion in the Pantheon of Rome”. In: *Eng Struct* 164 (2018), pp. 259–273. DOI: [10.1016/j.engstruct.2018.02.082](https://doi.org/10.1016/j.engstruct.2018.02.082).
- [16] **F Masi**, I Stefanou, and P Vannucci. “On the origin of the cracks in the dome of the Pantheon in Rome”. In: *Eng Fail Anal* 92 (2018), pp. 587–596. DOI: [10.1016/j.engfailanal.2018.06.013](https://doi.org/10.1016/j.engfailanal.2018.06.013).

5.2 Book chapters

- [A1] **F Masi**. “Introduction to regression methods”. In: *Machine Learning in Geomechanics 1: Overview of Machine Learning, Unsupervised Learning, Regression, Classification and Artificial Neural Networks*. Ed. by I Stefanou and F Darve. Wiley, 2024. ISBN: 978-1-394-32555-9. DOI: [10.1002/9781394325634.ch2](https://doi.org/10.1002/9781394325634.ch2).
- [A2] **F Masi** and I Stefanou. “Physics-informed and thermodynamics-based artificial neural networks for constitutive modeling”. In: *Machine Learning in Geomechanics 2: Data-Driven Modeling, Bayesian Inference, Physics- and Thermodynamics-based Artificial Neural Networks and Reinforcement Learning*. Ed. by I Stefanou and F Darve. Wiley, 2024. ISBN: 978-1-394-32564-1. DOI: [10.1002/9781394325665.ch3](https://doi.org/10.1002/9781394325665.ch3).

5.3 Reviewed international conferences

- [B1] **F Masi**, I Stefanou, et al. “Material modeling via thermodynamics-based artificial neural networks”. In: *Geometric Structures of Statistical Physics, Information Geometry, and Learning*. Ed. by F Barbaresco and F Nielsen. Les Houches, France: Springer International Publishing, 2021, pp. 308–329. DOI: [10.1007/978-3-030-77957-3_16](https://doi.org/10.1007/978-3-030-77957-3_16).
- [B2] **F Masi**, I Stefanou, et al. “Response of monumental buildings to internal explosions”. In: *7th ECCOMAS Thematic Conference on Computational Methods in Structural Dynamics and Earthquake Engineering*. Ed. by M Papadrakakis and M Fraiadakis. Crete, Greece (short version of [15]), 2019. DOI: [10.7712/120119.6958.19630](https://doi.org/10.7712/120119.6958.19630).
- [B3] **F Masi**, I Stefanou, et al. “Rocking response and overturning of museum artefacts due to blast loading (invited keynote)”. In: *7th ECCOMAS Thematic Conference on Computational Methods in Structural Dynamics and Earthquake Engineering*. Ed. by M Papadrakakis and M Fraiadakis. Crete, Greece (application of the theory developed in [12]), 2019. DOI: [10.7712/120119.7119.19577](https://doi.org/10.7712/120119.7119.19577).

5.4 Other international publications

- [C1] A Morsel, **F Masi**, et al. “Reduced-scale testing of masonry structures to explosions (poster)”. In: *34th Workshop ALERT Geomaterials*. Aussois, France, 2023.
- [C2] A Morsel, **F Masi**, et al. “Design of reduced-scale experiments of masonry structures subjected to blast loads (poster)”. In: *33rd Workshop ALERT Geomaterials*. Aussois, France, 2022.
- [C3] **F Masi** and I Stefanou. “Thermodynamics-based Neural Networks: a general framework for modeling microstructured materials displaying path-dependency (poster)”. In: *32nd Workshop ALERT Geomaterials*. Aussois, France (dissemination of [7]), 2021.
- [C4] F Rabie, **F Masi**, et al. “Thermodynamics-based Artificial Neural Networks for Nonlinear Seismic Analysis of High-rise Buildings (poster)”. In: *32nd Workshop ALERT Geomaterials*. Aussois, France, 2021.
- [C5] **F Masi**, I Stefanou, et al. “A Discrete Element Method approach for the preservation of the architectural heritage against explosions”. In: *12th HSTAM International Congress on Mechanics*. Thessaloniki, Greece, 2019.
- [C6] **F Masi**, I Stefanou, et al. “Michelangelo’s David or Aphrodite of Milos: Who is More Resistant to Blast Loads?” In: *12th HSTAM International Congress on Mechanics*. Thessaloniki, Greece (preliminary, short version of [10]), 2019.

5.5 Reviewed national conferences

- [D1] **F Masi** and I Stefanou. “Réseaux de neurones artificiels basés sur la thermodynamique (TANN) pour la mécanique computationnelle et la modélisation multi-échelle”. In: *25^e Congrès Français de Mécanique*. Nantes, France (short version of [7] submitted as application to AFM’s Paul Germain Prize), 2022.

- [D2] **F Masi**, I Stefanou, et al. "Reduced-scaled experiments of masonry structures under blast loads". In: *25^e Congrès Français de Mécanique*. Nantes, France (dissemination of [9]), 2022.
- [D3] G Piuino, **F Masi**, et al. "Multi-scale modelling of natural composites via Thermodynamics-based Artificial Neural Networks". In: *25^e Congrès Français de Mécanique*. Nantes, France, 2022.

5.6 Research reports

- [E1] **F Masi**. "Fast-dynamic response and failure of masonry structures of non-standard geometry subjected to blast loads". PhD thesis. École centrale de Nantes, 2020.
- [E2] P Vannucci, I Stefanou, et al. *Structural integrity of Notre Dame Cathedral after the fire of April 15th, 2019*. Tech. rep. Paris: UVSQ-ENPC, 2019.
- [E3] P Vannucci, **F Masi**, et al. *A comparative study on the effects of blast actions on a monumental structure*. Tech. rep. Paris: UVSQ-ENPC, 2017.
- [E4] P Vannucci, I Stefanou, et al. *Cathédrales Durables (classified: Confidentiel Défense)*. Tech. rep. Paris: CNRS, 2017.

5.7 Invited talks and seminars

- Invited lecture, "Data and thermodynamics are all you need: constitutive model discovery under hard constraints," *Numerical Methods in Geotechnics*, Technische Universität Hamburg, Hamburg, Germany, 17–18 September 2026.
- Invited lecture, "Thermodynamics as a hard constraint: learning constitutive models from stress–strain data," *AI in Mechanics* scientific seminar, Speinshart Scientific Center for AI and SuperTech, Speinshart, Germany, 17–20 May 2026.
- Invited lecture, "Discovering constitutive models from data and physics via hard constraints," *MePhy & I-Gaia Day: Machine Learning in Mechanics and Physics*, Paris, France, 9 December 2025.
- Invited lecture, "Réseaux de neurones artificiels basés sur la thermodynamique et modélisation multi-échelle," *Mécatat National Congress*, Aussois, France, 29 January 2025.
- Invited lecture, "Data-driven constitutive equations for complex materials and where to find them," *Australian Geotech Webinars*, Australia, 4 October 2024.
- Invited lecture, "Discovery of neural constitutive equations of complex materials," *3SR Laboratory*, Université Grenoble Alpes, Grenoble, France, 5 October 2023.
- Invited lecture, "Physics-based neural constitutive equations," *Hybrid Computational Methods in Geotechnics*, The University of Melbourne, Australia, 27 April 2023.
- Invited lecture, "À la (re-)découverte des lois de comportement par la thermodynamique et l'intelligence artificielle," *IA: Approches et intérêt pour l'étude des CMC – Atelier, Groupement de Recherche (CMC)²*, ENS Paris-Saclay, France, 17 March 2023.
- Invited lecture, "Thermodynamics-based Artificial Neural Networks," *Inria TAU team*, Paris-Saclay, France, 28 February 2023.
- Invited lecture, "Data- and thermodynamics-driven discovery of constitutive equations," *ISSMGE TC309 Technical Forum of Young Scholars on Data-driven Modelling of Soil Behaviours with Geotechnical Applications*, Hong Kong Polytechnic University, 25 November 2022.
- Invited lecture, "Machine learning meets thermodynamics," *Navier Laboratory*, École des Ponts ParisTech, Marne-la-Vallée, France, 14 October 2022.
- Invited workshop contribution, "Deep learning, simulation temps réel et réduction de modèles," *5^e Workshop of the Computational Structural Mechanics Association (CSMA) Junior*, Giens, France, 14–16 May 2022.
- Invited lecture, "How can Machine Learning help in earthquake control and fault mechanics?," *Crunch Machine Learning + X Seminars*, Brown University, Division of Applied Mathematics, 12 November 2021 ([video](#)).
- Invited lecture, "Can we tame earthquakes?," *Data-Centric Engineering*, The University of Sydney, 27 October 2021.

6 Supervision and teaching

6.1 Supervision

– Postdoctoral researchers

- Edoardo Fabbrini (05/2026 – present). Postdoctoral fellow in the MIAI Cluster IA Chair [AIM](#), in collaboration with Vincent Acary. Subject: scientific computing and machine learning for computational mechanics.

– PhD students

- Henri Leroy (11/2025 – present). PhD student at Université Grenoble Alpes (LJK/Inria), in the MIAI Cluster IA Chair [AIM](#), co-supervised with Vincent Acary. Subject: physics-informed and data-driven modelling and scientific computing for granular materials.
- Florian Vincent (10/2023 – present). PhD candidate at Université Grenoble Alpes (LJK/Inria), co-supervised with Vincent Acary and Jérôme Malick. Subject: robust optimization and implicit constitutive neural models.

- Ahmad Morsel (01/2021 – 04/2024) – PhD at École Centrale de Nantes, in the frame of BLAST ([Blast LoAds on Structures](#)) – Connect Talent project, funded by Pays de la Loire and Nantes Metropole, in collaboration with Ioannis Stefanou and Panagiotis Kotronis. Subject: Experimental testing of masonry structures subjected to extreme loads.

– Master students

- Enzo Louvard (02/2024 – 07/2024) – Master internship at The University of Sydney. Subject: Neural constitutive equations for material model discovery from small data. Amount of supervision: 80% (in collaboration with Itai Einav).
- Farah Rabie (02/2021 – 08/2021) – Master internship at École Centrale de Nantes, in the frame of CoQuake ([Controlling earthQuakes](#)) – European Research Council (ERC) Starting Grant. Subject: Thermodynamics-based Artificial Neural Networks for nonlinear seismic analysis of high-rise buildings. Amount of supervision: 60% (in collaboration with Ioannis Stefanou).
- Barbara Abougaye (02/2020 – 08/2020). Master internship at École Centrale de Nantes. Subject: Scaling laws for the material response under impact loading. Amount of supervision: 60% (in collaboration with Ioannis Stefanou).

6.2 Teaching

- Université Grenoble Alpes (2025–present).
 - “Machine Learning for Physics” (2026–present), Master MSIAM/ENSIMAG – course coordinator and lecturer.
 - “Machine Learning for Mechanics” (2026–present), Master MGCEM – course coordinator and lecturer.
 - “Introduction to Numerical Methods” (2025), Bachelor L3M/L3MAA – lecturer.
- ALERT Geomaterials Olek Zienkiewicz School on Constitutive Modelling of Geomaterials (2025), Charles University, Prague. “Introduction to Machine Learning” and “Constitutive Modelling Meets Machine Learning: Theory and Applications,” with a hands-on session, [programme](#).
- ALERT Geomaterials Doctoral School on Machine Learning in Geomechanics (2023). “Introduction to Regression Methods” and “Physics-Informed and Thermodynamics-Based Neural Networks” ([lecture video](#)), combining theory, hands-on examples, and exercises; [course material](#).
- Computational Structural Mechanics Association Junior Workshop (2022). “Deep learning, simulation temps réel et réduction de modèles,” with hands-on material and exercises.
- École Centrale de Nantes (2020–2023). “Experimental Imaging Analysis for Engineers” (Master, course coordinator and lecturer) and “Continuum Mechanics” (Bachelor, lecturer). Responsibilities included lectures, problem classes, exam preparation, and grading.
- École nationale des ponts et chaussées (2018–2019). “Advanced Computational Mechanics” (Master) and “Computational Mechanics” (Bachelor), lecturer. Responsibilities included problem classes, midterm grading, and team-project design.

7 Service

7.1 Responsibilities

- Invited reviewer for: Computational Mechanics – Computer Methods in Applied Mechanics and Engineering – Computer and Geotechnics - Experimental Mechanics - Defense Technology - Géotechnique – Internal Journal of Mechanical Sciences - International Journal for Numerical and Analytical Methods in Geomechanics – International Journal for Numerical Methods in Engineering – Mechanics of Materials - Journal of the Mechanics and Physics of Solids – Reliability Engineering & System Safety - Soil Dynamics and Earthquake Engineering.
- Invited reviewer in national funding schemes.
- Co-organizer of the Minisymposium entitled [Scientific Machine Learning techniques for complex engineering systems](#) on the occasion of the 2023 ECCOMAS Young Investigators Conference ([YIC2023](#)), held in Porto, Portugal (19-21 June, 2023).
- Member of the [Direction Board of GeM laboratory](#), École Centrale de Nantes (2022).
- Committee member of [Australian Association for GRAnular Media \(ANAGRAM\)](#) (11/2023 – present).

7.2 Dissemination of scientific knowledge

- Co-coordinator of the “[Explosions, vibrations et patrimoine](#)” workshop (with Ahmad Morsel, Ioannis Stefanou, Panagiotis Kotronis, Guillaume Racineux, Emmanuel Marché), on the occasion of the 2023 edition of the *Nuit Blanche des Chercheur.es* event hosted by Nantes Université ([short video](#)).
- [Blog on my PhD experience](#) in the form of an interview aimed at younger audiences about academia, industry, and international exchanges, published by [Docteurs Sciences pour l'Ingénieur](#), in July 2022.
- Ambassador of [DECLICS](#) Association: *Dialogues Entre Chercheurs et Lycéens pour les Intéresser à la Construction des Savoirs*, 2020-2022.
- Webinar for a wide professional (civil engineering) public: “[Comportement des structures maçonnées à l'explosion](#),” Ingérop, 29 October, 2020.